Answer true to the correct statement and False to the incorrect statement and correct it .

1. ( ) PID controller is more stable than PI, PD controller .

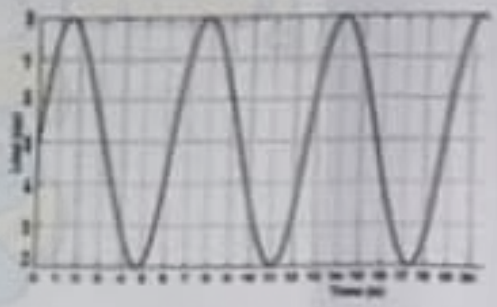
2. ( ) calibration diagram of the pump is the graph between the flow and voltage and it can be nonlinear of proportionality .

3. ( ) the time constant of the first order control system is equal to 0.63 of the value of the level

4. ( ) time response is the interval between input step and output signal first reaching of its final value of the level .

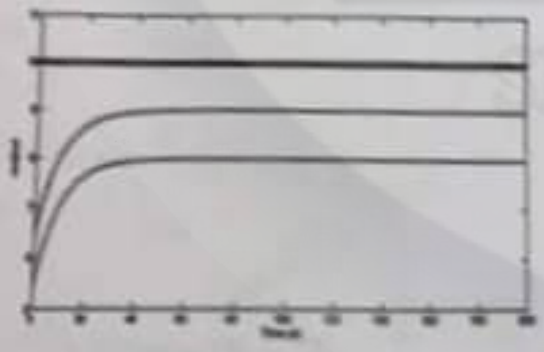
5. ( ) The derivative controller reduces the amount of over shoot but increase delay time .

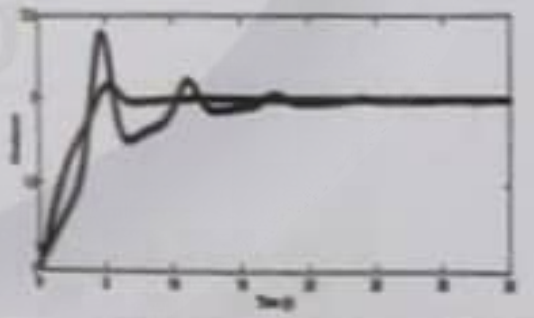
QUESTION 3: A- The following graph is obtained from ON-OFF control of the level experiment. Answer the following questions:



1. The Set point is=
2. On time is=
3. Off time is=
4. Experimental value of hysteresis is =
5. Draw the On-Off time in the graph
6. Draw the closed loop control of the process for ON-OFF controller. Define each block its gain.

B. Given the following seven controllers (PD1010, I 20, PID 10105, D10,PD3010, PID20202, ID1010,). Assign each controller with the appropriategraph explaining why.





a) P 10 b) PI 10, 10 c) PID 10,10, 10 e) _____

10) Approximately the overshoot percentage for system 1 , system 2 =

a) 0 , 7% b) undefined , 70% c) undefined , 78.5% e) _____

11) Which of the following controller is suitable for the system 2

a) PD 10, 10 b) PD10, 20 c) P 10 e) _____

12) The theoretical error for system 1 , system 2

a) 0, 0 b) A/ 1+KPGPR , 0 c) 1.2 , 0 e) _____

13) Which of the following controller gain is suitable for the system 1

a) kp+kds b) kp+ kds+ki/s c) kp+kis e) _____

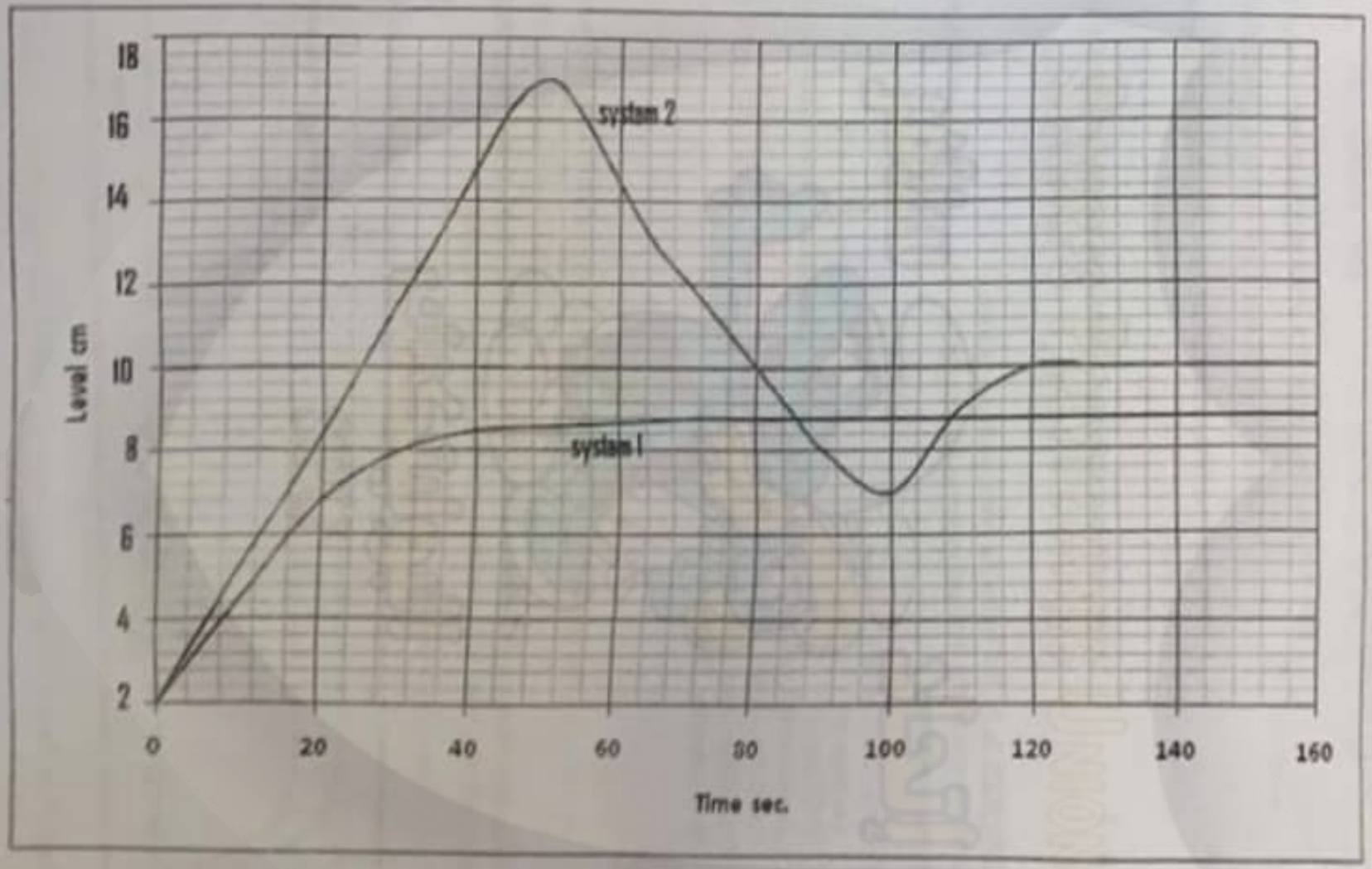
14) Approximately the overshoot for system 1 , system 2 =

a) 0 , 7 b) undefined , 7 c) undefined , 17 e) _____

QUESTION 2:

A) - If we decrease the value of the integral term for PI (4, 2) what changes will occur for the following variables , use the words (increase, decrease, the same)

1) Tss
2) Over shoot
3) Number of cycles in settling time
4) steady state error
5) stability of the system


18
16
14
12
10
8
6
4
2
Level cm
system 2
system I
0
20
40
60
80
100
120
140
160
Time sec.

Al Balqa Applied University, Faculty of Engineering Technology
Mechanical Engineering Department, 1 st semester 2022/2023

mark: / 50

Mechanical control lab

student's name :.....
Day & time of the lab:.....

QUESTION 1: circle the right answer :

1) The steady state level (hss) for system 1 , system 2 =

a) 6.8 , 8 b) 8.8 , 10 c) 6.8 , 14 e) _____

2) Approximately Time constant for system 1 , system 2 =

a) 12, 15 b) 25 , 30 c) 16 , 17 e) _____

3) the experimental error ess for system 1 , system 2 =

a) 3.6 , 0 b) 1.2 , 0 c) 1.6 , 0 e) _____

4) Approximately the settling time for system 1 , system 2 =

a) 0 , 128 b) undefined , 128 c) 64 , 128 e) _____

5) Approximately the rise time for system 1 =

a) 30 b) 20 c) 60 e) _____

6) Approximately the rise time for system 2 =

a) 30 b) 20 c) 25 e) _____

7) Approximately the delay time for system 1 =

a) 5.4 b) 20 c) 15 e) _____

8) Approximately the delay time for system 2 =

a) 5.4 b) 12 c) 6 e) _____

9) Which of the following controller is suitable for the system 1

Date of Exam: 03-09-2017 Saturday

QUESTION 1: (10 Marks)

1. What is the function Of LVDT?

measure the displacement and convert it into voltage.

2. What are the disadvantages of the ON – OFF controller?

1- ~~High~~ Power consumption / electricity consumption

2- Physical damage on pumps, due rapidly swithing

3. Explain how you can calculate the Resistance of the drain valve experimentally?

we must Know if the flow is luminar or turbilant.

$R = 2H/Q$

$$\frac{\text{flow velocity} * D_{Pipe}}{1 * 10^{-6}\ m^2/s}$$

4. **Define**

a. Time constant time @ 0.63 hss+hi

main characteristic unit of the first order.

b. Rising time time @ 0.9*hss+hi - time @ 0.1*hss+hi

the time needed to reach the ~~level~~ responce. to maximum value. to reach

c. Delay time time @ 0.5*hss+hi

time needed for the responce to reach half of its final value

d. Settling time the time when the level start to be constan

(no ~~Change~~ Change in level)

3] a-This graph was obtained for on-off control of the level Exp#5, answer the following:



1-The upper limit =

2- The lower limit =

1-The set point =

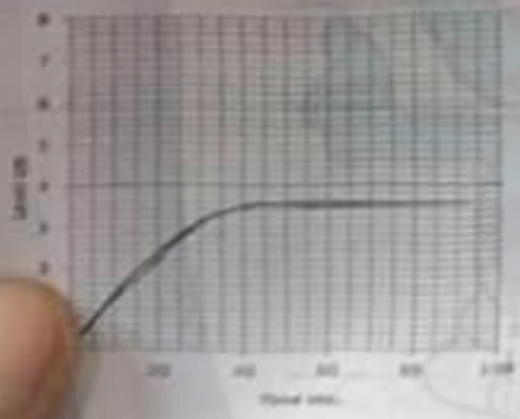
4- time ON =

5- time OFF =

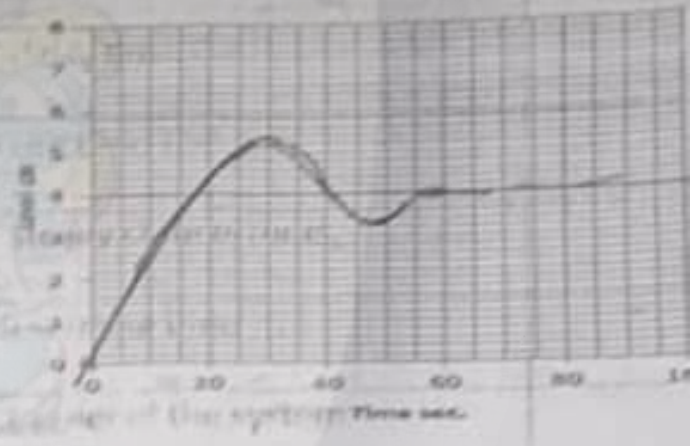
6- hysteresis =

Q4) Given the following seven controllers (P10 , I10 , PI 105, D5 , P20 , PI 1010 , ID 1010)

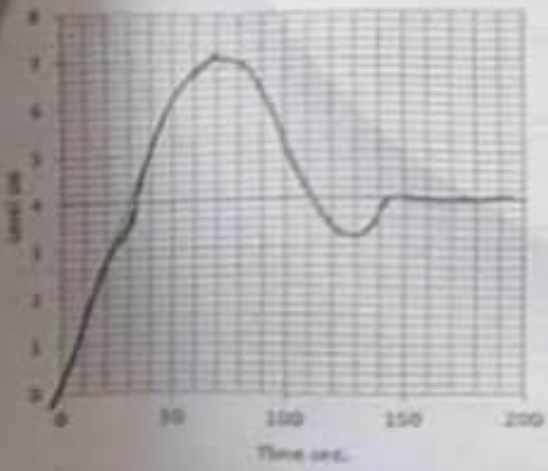
Assign each controller with the appropriate case graph and explain why? (8 marks)



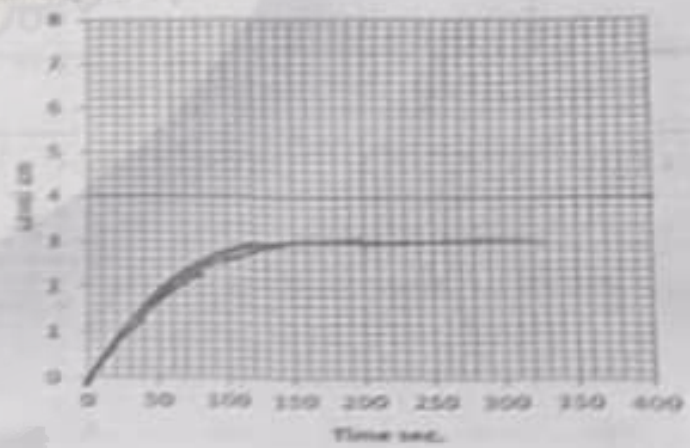
Case 1:



Case 2:



Case 3:



Case 4:

Al Balqa Applied University,
Faculty of Engineering Technology
Mechanical Engineering Department

Mechanical control lab

Date :

Name : ....
Day & time of the lab : .....

Q1) Answer **True** or **False** for the following statements and **correct the false** : (10 m

1- Calibration the LVDT below 14 cm because above this level the relation is non linear.( )

2- In exp. 6 on-off control of the level with sol valve, the sol valve give signal to the pump to on or off according to the tank level.( )

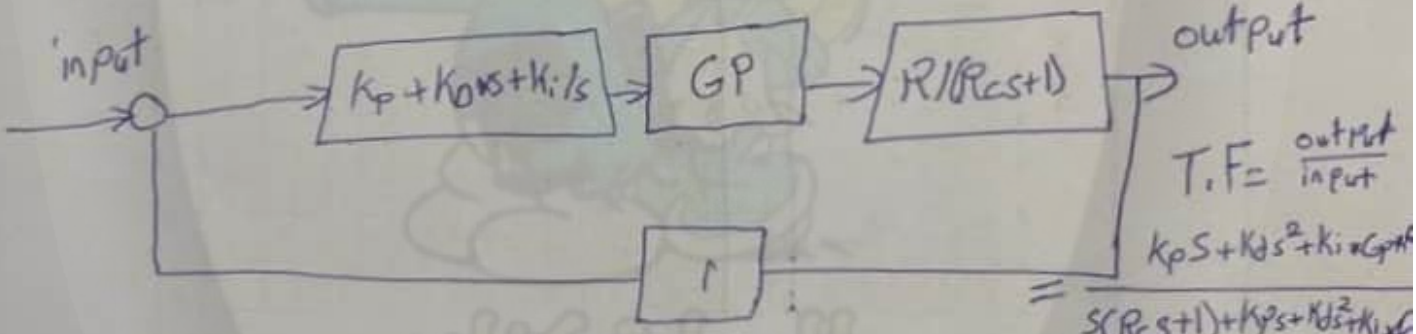
3- In exp. 5 on –off control of the level if the upper limit in the tank = 10 cm and the lower limit = 4cm then the value of the hysteresis = 40%. ( )

4- A-In exp. 4 if the tank diameter = 17.5 cm and the static gain = 1.25 s/cm^2 then the time constant of the process = 904 sec. ( )

b- If the level in the tank H= 15 cm , then the flow velocity of the water exit from the drain = 1.54m/s ( )

Peak Time T_p	4.9	50
Delay Timt T_d	time@ .5(hmax - hi) + hi = 3 sec	5 sec
Steady State Error e_{ss}	setpoint - ho = 3 - 3 = 0	11 - 2.5 = 8.5
Settling Time	25	50
System Order	second order	first order
Controller Transfer Function	$\frac{Kps + Kds^2 + KI \times Gp \times R}{S(Rcs+1) + Kps + Kds^2 + Ki \times Gp R}$	$\frac{(Kp + Kds) \times Gp \times (\frac{R}{Rcs+1})}{1 + (Kp + Kds) \times Gp \times (\frac{R}{Rcs+1}) \times G}$
Theoretical error	0	$\frac{\text{step input (A)}}{1 + Kp\ Gp\ R}$
Time Constant	time@ 0.63(hmax - hi) + hi = 3.75 sec	16 sec
Experimental Value of R	$R = \frac{A}{5}$	$R = \frac{A}{5}$
Rise Time	3.8 sec	50 sec

B. Draw the closed loop control of the process for the **PID** controller. Define each block its gain. Write the transfer function.



T.F = output / input

$= \frac{Kps + Kds^2 + Ki \times Gp \times R}{S(Rcs+1) + Kps + Kds^2 + Ki \times Gp \times R}$

C. Draw the response of System 1 on the same graph if the gain of Derivative controller is 2 explaining why??

D. What are the objective of these experiments?

to verify the effects of the gain of the loop (Kp)(Kd)(Ki) on the dynamic response of the system.

QUESTION 3: (8 Marks)

a. For a closed loop system using on off controller, the desired set point = 4 cm and the hysteresis = 30%. What are the values of the upper and lower limits?

upper limit = setpoint + $\frac{1}{2}$ hysteresis $\times 10$

$= 4 + (15 \times 10) = 34$

lower limit = setpoint - $\frac{1}{2}$ hysteresis $\times 10$

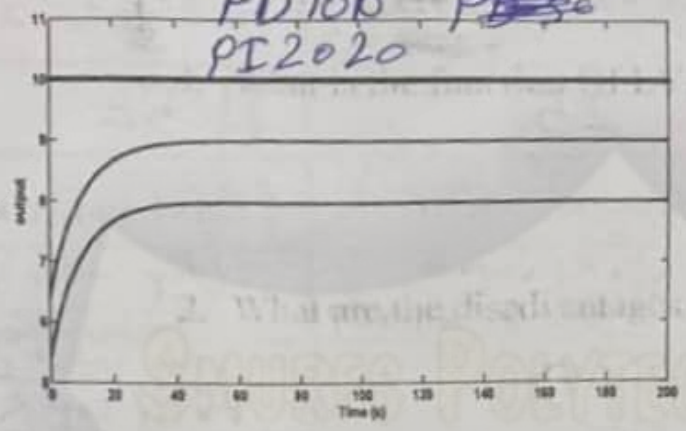
$= 4 - (15 \times 10) = -26$

b. Given the following seven controllers **(PD1010, I 20, PID 10105, D10, PD3010, PI2020, ID1010,).** Assign each controller with the appropriate graph explaining why.

PD 3010

PD 1010

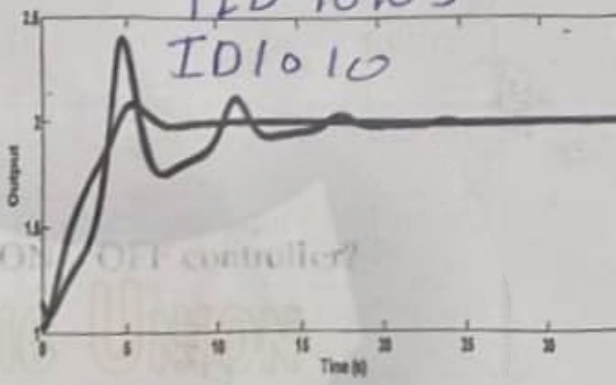
PI 2020



D 10

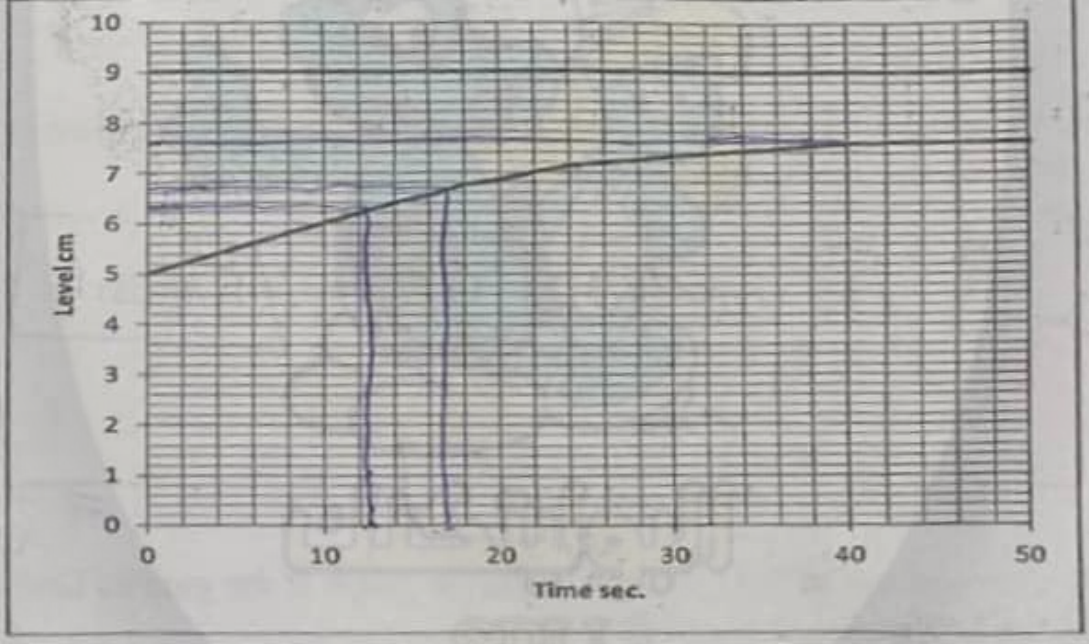
PID 10105

ID 1010



QUESTION 4: (8 Marks)

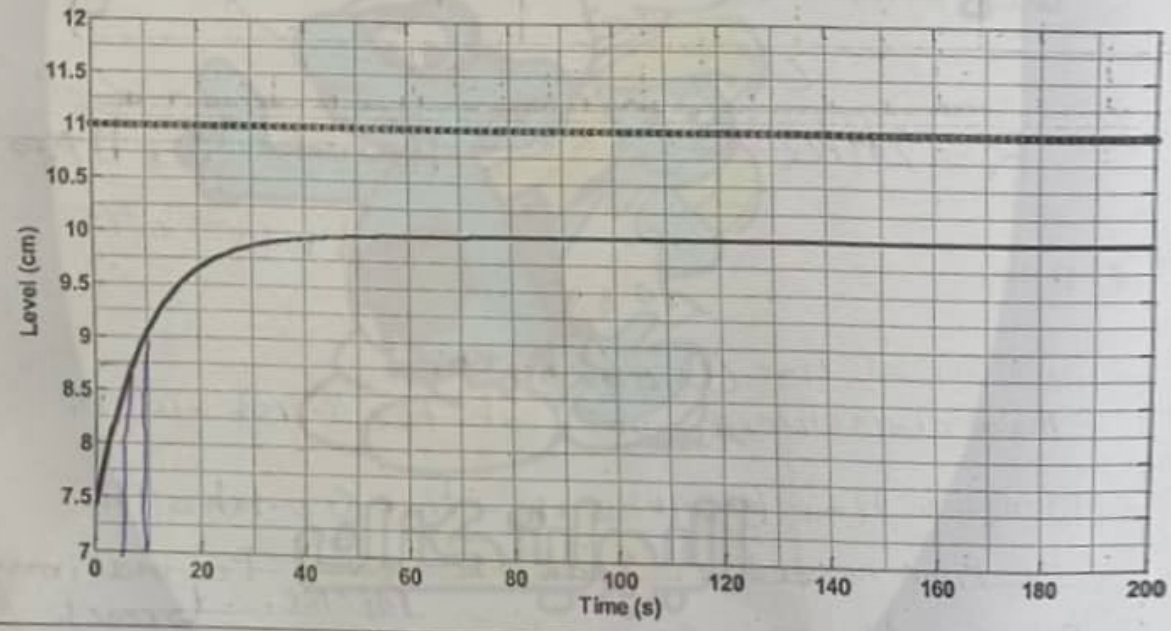
The given graph represents the transient response of the closed loop proportional control level K_p =5 . Use the graph to fill the table (10 marks)



1-The initial condition= 5	**5- Steady state level h_{ss} =** 7.6−5=2.6
2- delay time T_d .5*(hmax−hi) +hi = 12.5 sec	**6-Time constant of the process τ** .65*(hmax−hi) +hi = 17 sec
3- the step input value 2	**7-steady state error e_{ss}** 9−7.6=1.4
4-The desired value = 7.6	**8-settiling time T_{ss}** 40s

QUESTION 2: the following figure represents the transient response of the closed loop **PID 1010** and **PD 2010** control of the level. Use the graph to fill the table, show all the projections. The Diameter of the cylinder is 17.5 cm. **(22 Marks)**





	SYSTEM 1	SYSTEM 2
Steady State Level h_{ss}	3.4 - 2.1 = 1.3	10 - 7.5 = 2.5
Step input	~~5~~ ~~2.5~~ 2.5	~~20~~ 10
Maximum Over shoot	0.4	undifined
Minimum Overshoot	S.P - h min 3 - 2.75 ~~undifined~~ = 0.25	undifined
Maximum Over shoot Percentage	$\frac{0.5}{S.P} \times 100\% = \frac{0.4}{3} = 13.3\%$	0
Minimum Over shoot Percentage	$\frac{0.5}{S.P} \times 100\% = \frac{0.25}{3}$ $= 8.33\%$	0

Al Balqa Applied University,
Faculty of Engineering Technology
Mechanical Engineering Department

Mechanical control lab

Date :

Name : ....
Day & time of the lab : .....

Q1) Answer **True** or **False** for the following statements and **correct the false** : (10 m

1- Calibration the LVDT below 14 cm because above this level the relation is non linear.( )

2- In exp. 6 on-off control of the level with sol valve, the sol valve give signal to the pump to on or off according to the tank level.( )

3- In exp. 5 on –off control of the level if the upper limit in the tank = 10 cm and the lower limit = 4cm then the value of the hysteresis = 40%. ( )

4- A-In exp. 4 if the tank diameter = 17.5 cm and the static gain = 1.25 s/cm^2 then the time constant of the process = 904 sec. ( )

b- If the level in the tank H= 15 cm , then the flow velocity of the water exit from the drain = 1.54m/s ( )

3) a-This graph was obtained for on-off control of the level Exp#5, answer the following:



1-The upper limit =

2- The lower limit =

1-The set point =

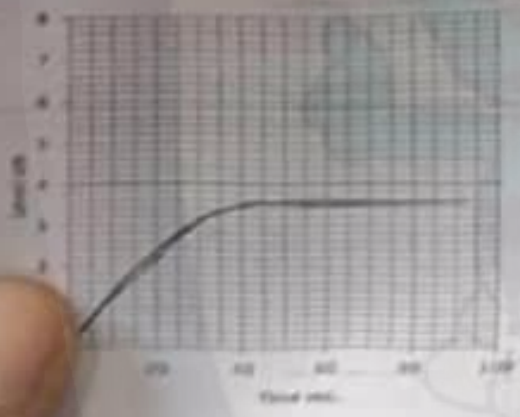
4- time ON =

5- time OFF =

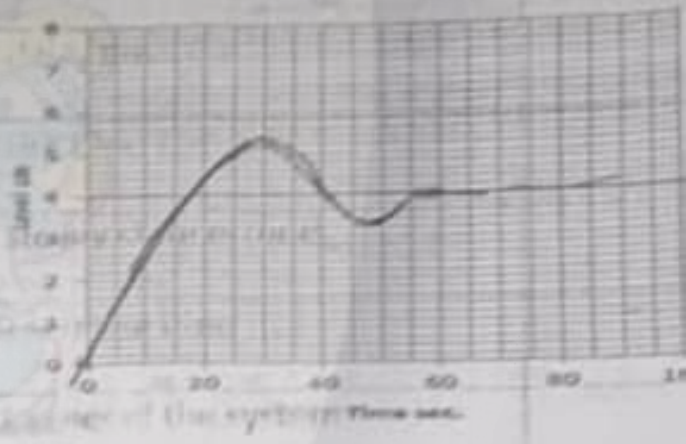
6- hysteresis =

Q4) Given the following seven controllers (P10 , I10 , PI 105, D5 , P20 , PI 1010 , ID 1010)

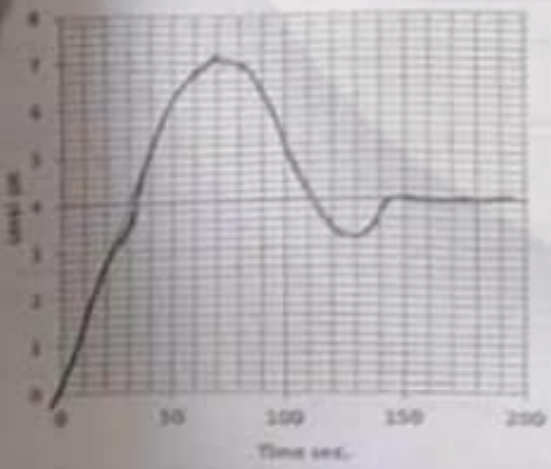
Assign each controller with the appropriate case graph and explain why? (8 marks)



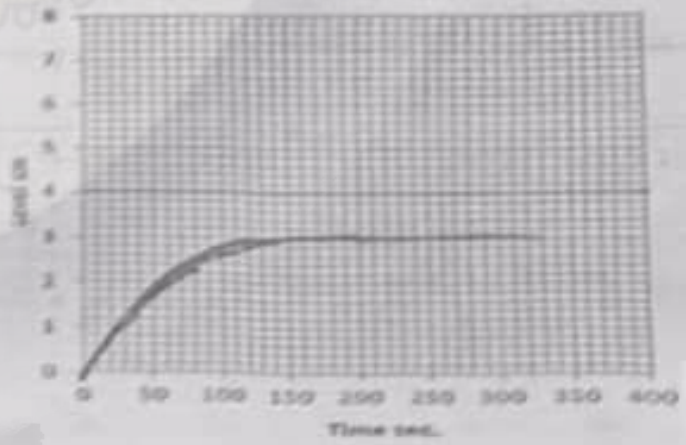
Case 1:



Case 2:



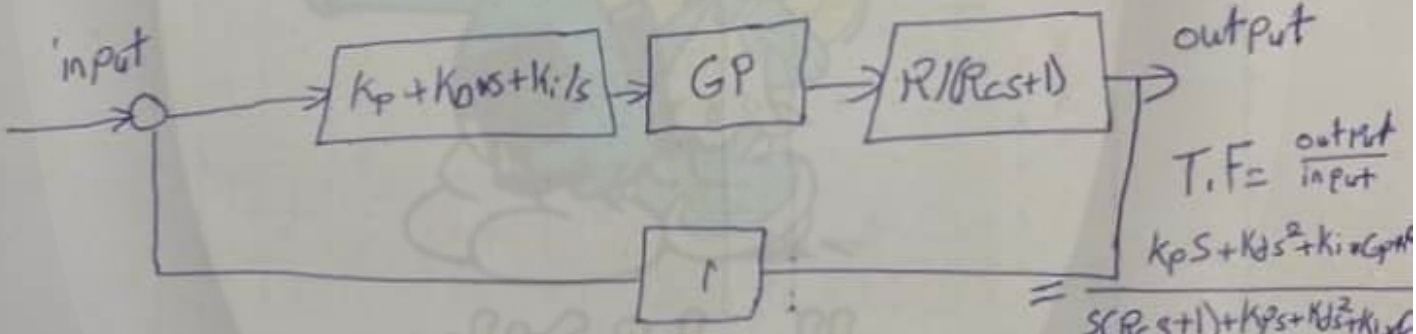
Case 3:



Case 4:

Peak Time T_p	4.9	50
Delay Timt T_d	time@ .5(hmax - hi) + hi = 3 sec	5 sec
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Theoretical error	0	$\frac{\text{step input (A)}}{1 + Kp\ Gp\ R}$
Time Constant	time@ 0.63(hmax - hi) + hi = 3.75 sec	16 sec
Experimental Value of R	$R = \frac{A}{5}$	$R = \frac{A}{5}$
Rise Time	3.8 sec	50 sec

B. Draw the closed loop control of the process for the **PID** controller. Define each block its gain. Write the transfer function.



$T.F = \frac{\text{output}}{\text{input}} = \frac{Kps + Kds^2 + Ki \times Gp \times R}{s(Rcs+1) + Kps + Kds^2 + Ki \times Gp \times R}$

C. Draw the response of System 1 on the same graph if the gain of Derivative controller is 2 explaining why??

D. What are the objective of these experiments?

to verify the effects of the gain of the loop (Kp)(KD)(Ki) on the dynamic response of the system.

QUESTION 3: (8 Marks)

a. For a closed loop system using on off controller, the desired set point = 4 cm and the hysteresis = 30%. What are the values of the upper and lower limits?

upper limit = setpoint + $\frac{1}{2}$ hysteresis $\times 10$

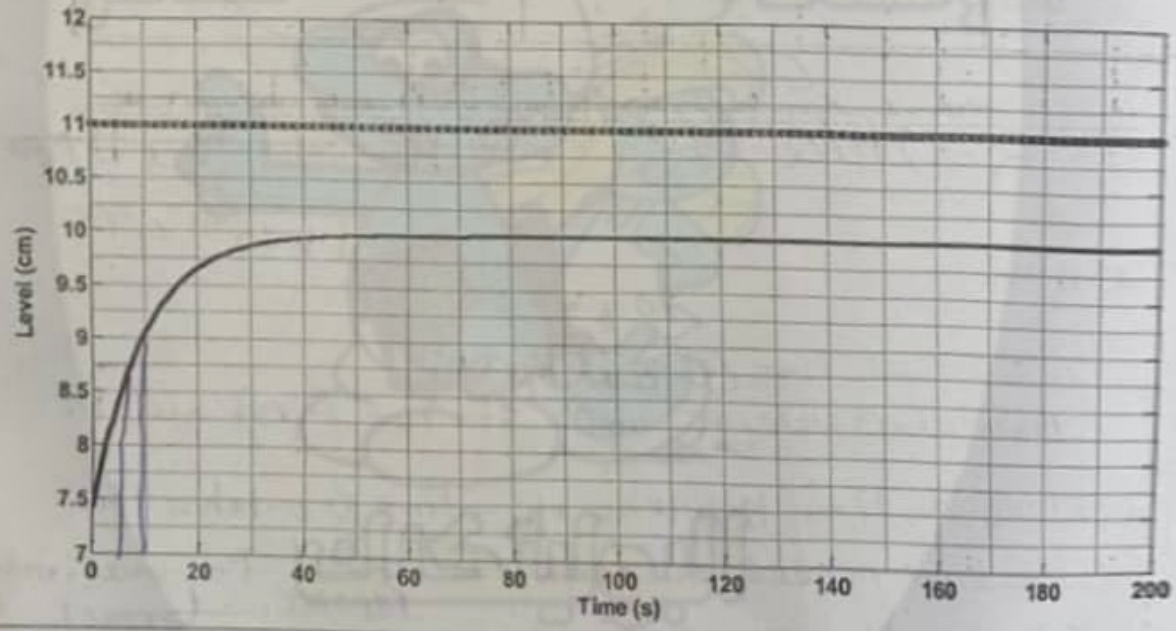
$= 4 + (15 \times 10) = 34$

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$= 4 - (15 \times 10) = -26$

QUESTION 2: the following figure represents the transient response of the closed loop **PID 1010** and **PD 2010** control of the level. Use the graph to fill the table, show all the projections. The Diameter of the cylinder is 17.5 cm. **(22 Marks)**





	SYSTEM 1	SYSTEM 2
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Step input	~~5~~ ~~2.5~~ 2.5	~~20~~ 10
Maximum Over shoot	0.4	undifined
Minimum Overshoot	S.P - h min 3-2.75 =0.25	undifined
Maximum Over shoot Percentage	$\frac{0.5}{S.P} \times 100\% = \frac{0.4}{3} = 13.3\%$	~~*~~ 0
Minimum Over shoot Percentage	$\frac{0.5}{S.P} \times 100\% = \frac{0.25}{3}$	0

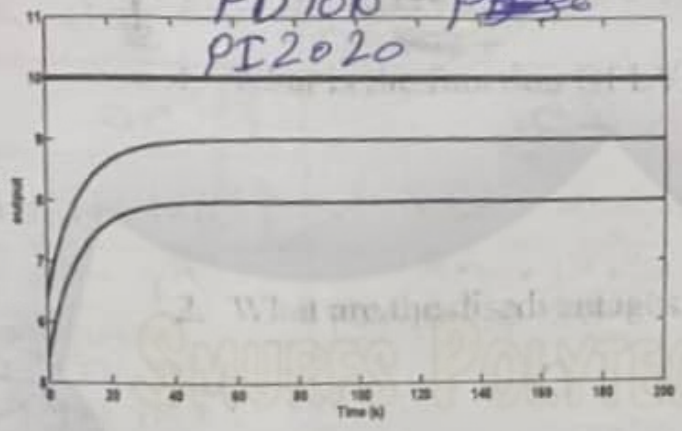
$= 8.33\%$

b. Given the following seven controllers **(PD1010, I 20, PID 10105, D10, PD3010, PI2020, ID1010,).** Assign each controller with the appropriate graph explaining why.

PD 3010

PD 1010

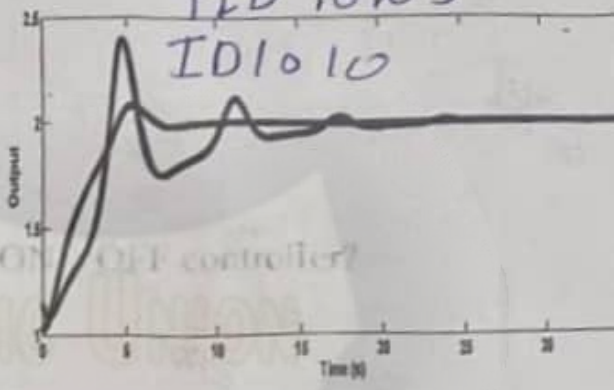
PI 2020



D 10

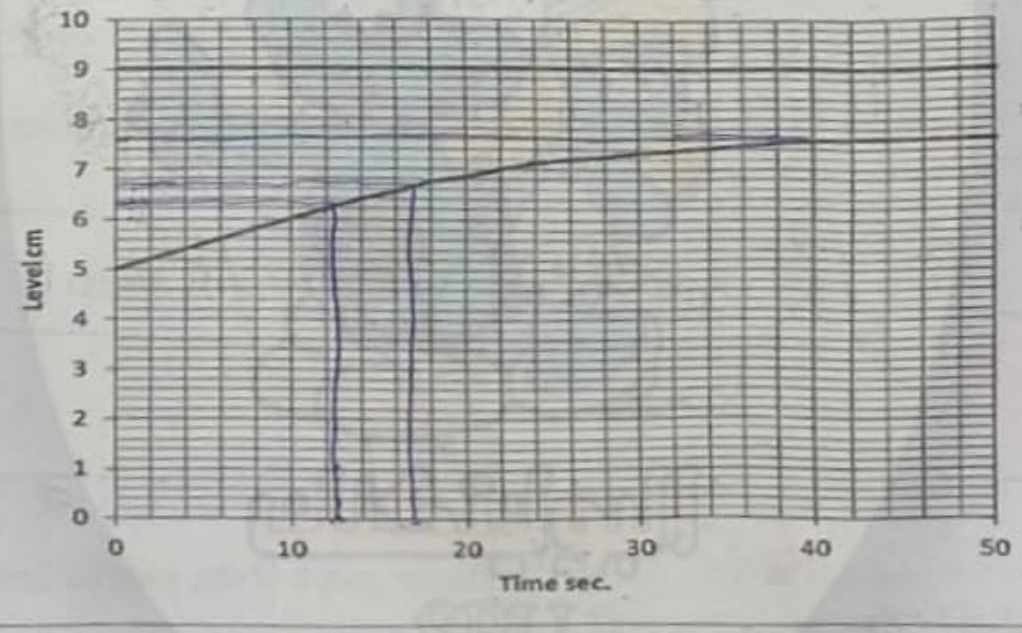
PID 10105

ID 1010



QUESTION 4: (8 Marks)

The given graph represents the transient response of the closed loop proportional control level K_p =5 . Use the graph to fill the table (10 marks)



1-The initial condition= 5	**5- Steady state level h_{ss} =** 7.6-5=2.6
2- delay time T_d $.5*(h_{max}-h_i)+h_i$ = 12.5 sec	**6-Time constant of the process τ** $.65*(h_{max}-h_i)+h_i$ = 17 sec
3- the step input value 2	**7-steady state error e_{ss}** 9-7.6=1.4
4-The desired value = 7.6	**8-settiling time T_{ss}** 40s